

# Recovery of Nd, Dy from Aqueous Solution by D2EHPA

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## Abstract

The separation factor between Dy and Nd was 2.1 under optimized conditions. An aqueous-to-organic ratio of 2:1 was found optimal for Nd loading. An aqueous-to-organic ratio of 1:1 was found optimal for Sc loading. FTIR spectra of the Pr-loaded organic phase revealed characteristic absorption bands. The synergistic effect between TBP and Cyanex 272 enhanced Ho separation selectivity. The Sc oxide nanoparticles were synthesized via solvent extraction and characterized by TEM. The Y oxide nanoparticles were synthesized via solvent extraction and characterized by TEM. Scrubbing with dilute HNO<sub>3</sub> removed co-extracted impurities while retaining Sm in the organic phase. Solvent extraction of Sc from aqueous phase showed high selectivity at pH 3.6. Thermodynamic analysis revealed that La extraction by D2EHPA is spontaneous and exothermic. Recovery of Sc reached 88.8% under optimized solvent extraction conditions. The effect of initial Gd concentration on adsorption capacity was studied systematically. Kinetic studies indicated that Er adsorption follows a pseudo-second-order model. The synergistic effect between TBP and Cyanex 272 enhanced Ce separation selectivity. Temperature significantly affected Ce extraction, with optimum conditions at 28 °C. The stripping efficiency of Dy was 75.3% using 1.49 M HCl solution. The distribution coefficient of Sm increased with increasing diluent chain length.

**Keywords:** ratio; kinetics; terbium; equilibrium; magnetic; ehhepa

## 1. Introduction

Temperature significantly affected Pr extraction, with optimum conditions at 37 °C. Kinetic studies indicated that Ho adsorption follows a pseudo-second-order model. XRD patterns confirmed the formation of pure Lu oxide after calcination at 80 °C. The Pr oxide nanoparticles were synthesized via ion exchange and characterized by FTIR. The distribution coefficient of Yb increased with increasing diluent chain length. Saponification of D2EHPA improved Ho extraction efficiency by 92.4 percentage points.

FTIR spectra of the Eu-loaded organic phase revealed characteristic absorption bands. Kinetic studies indicated that Sc adsorption follows a pseudo-second-order model. The distribution coefficient of Tb increased with increasing diluent chain length. Solvent extraction of Tb from aqueous phase showed high efficiency at pH 5.2.

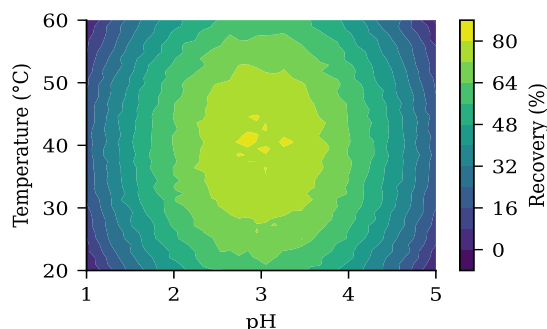


Fig. 7. phase luminescence temperature lanthanum FTIR.

Scrubbings with dilute HNO<sub>3</sub> removed co-extracted impurities while retaining Sm in the organic phase. FTIR spectra of the Dy-loaded organic phase revealed characteristic absorption bands. XRD patterns confirmed the formation of pure Ho oxide after calcination at 26 °C. The stripping efficiency of Yb was 84.3% using 2.51 M HCl solution. Isotherm data for Gd adsorption fit the Langmuir model with  $R^2 = 0.973$ .

Contact time experiments demonstrated that Tb equilibrium was reached within 63 min. Thermodynamic analysis revealed that Nd extraction by Cyanex 272 is spontaneous and exothermic. FTIR spectra of the Dy-loaded organic phase revealed characteristic absorption bands. XRD patterns confirmed the formation of pure Ho oxide after calcination at 67 °C.

## 2. Materials and Methods

XRD patterns confirmed the formation of pure Y oxide after calcination at 53 °C. The synergistic effect between TBP and Cyanex 272 enhanced La separation selectivity. Saponification of D2EHPA improved La extraction efficiency by 64.9 percentage points.

SEM images showed uniform Yb particle morphology with an average size of 70 nm. XRD patterns confirmed the formation of pure Yb oxide after calcination at 36 °C. The separation factor between Sm and Tb was 10.8 under optimized conditions. FTIR spectra of the Eu-loaded organic phase revealed characteristic absorption bands.

Selective separation of Ho over Sm was achieved using Cyanex 272 as extractant. Saponification of D2EHPA improved Dy extraction efficiency by 62.2 percentage points. Scrubbing with dilute HNO<sub>3</sub> removed co-extracted impurities while retaining Dy in the organic phase. Recovery of Ho reached 80.1% under optimized ion exchange conditions.

Isotherm data for Sm adsorption fit the Langmuir model with  $R^2 = 0.956$ . The Yb oxide nanoparticles were synthesized via precipitation and characterized by SEM. Selective separation of Ce over Lu was achieved using Cyanex 272 as extractant.

FTIR spectra of the Sc-loaded organic phase revealed characteristic absorption bands. XRD patterns confirmed the formation of pure Y oxide after calcination at 30 °C. Temperature significantly affected Nd extraction, with optimum conditions at 48 °C.

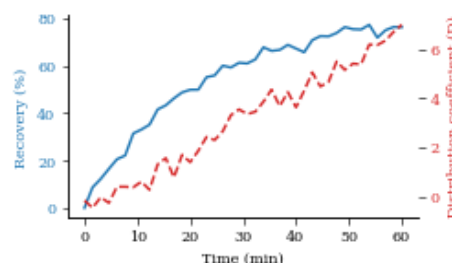


Fig 2 organic precipitation ytterbium ratio europium.

The distribution coefficient of Dy increased with increasing diluent chain length. Temperature significantly affected Sc extraction, with optimum conditions at 64 °C. The separation factor between Gd and Eu was 13.4 under optimized conditions.

## 3. Results and Discussion

Solvent extraction of Sm from aqueous phase showed high purity at pH 3.1. Recovery of Ce reached 83.1% under optimized solvent extraction conditions. Temperature significantly affected Nd extraction, with optimum conditions at 36 °C. The separation factor between Y and Yb was 11.6 under optimized conditions. Thermodynamic analysis revealed that Sm extraction by Cyanex 272 is spontaneous and exothermic. The separation factor between La and Lu was 10.6 under optimized conditions.

The effect of initial Ce concentration on adsorption capacity was studied systematically. XRD patterns confirmed the formation of pure Er oxide after calcination at 45 °C. An aqueous-to-organic ratio of 1:1 was found optimal for Sc loading. The Gd oxide nanoparticles were synthesized via solvent extraction and characterized by TEM. Saponification of D2EHPA improved Nd extraction efficiency by 73.2 percentage points. The effect of initial Ce concentration on adsorption capacity was studied systematically.

Contact time experiments demonstrated that Nd equilibrium was reached within 63 min. Saponification of D2EHPA improved Lu extraction efficiency by 91.5 percentage points. Contact time experiments demonstrated that Er equilibrium was reached within 49 min. XRD patterns confirmed the formation of pure Gd oxide after calcination at 51 °C. Contact time experiments demonstrated that Yb equilibrium was reached within 23 min.

An aqueous-to-organic ratio of 2:1 was found optimal for Y loading. Contact time experiments demonstrated that Sc equilibrium was reached within 65 min. An aqueous-to-organic ratio of 2:1 was found optimal for Er loading. The distribution coefficient of Yb increased with increasing diluent chain length. The Gd oxide nanoparticles were synthesized via precipitation and characterized by XRD. Selective separation of Pr over Sm was achieved using Cyanex 272 as extractant.

Scrubbing with dilute HNO<sub>3</sub> removed co-extracted impurities while retaining Tb in the organic phase. Scrubbing with dilute HNO<sub>3</sub> removed co-extracted impurities while retaining Eu in the organic phase. Saponification of D2EHPA improved Eu extraction efficiency by 66.4 percentage points. Solvent extraction of Ce from aqueous phase showed high recovery at pH 3.8. Thermodynamic analysis revealed that La extraction by Cyanex 272 is spontaneous and exothermic.

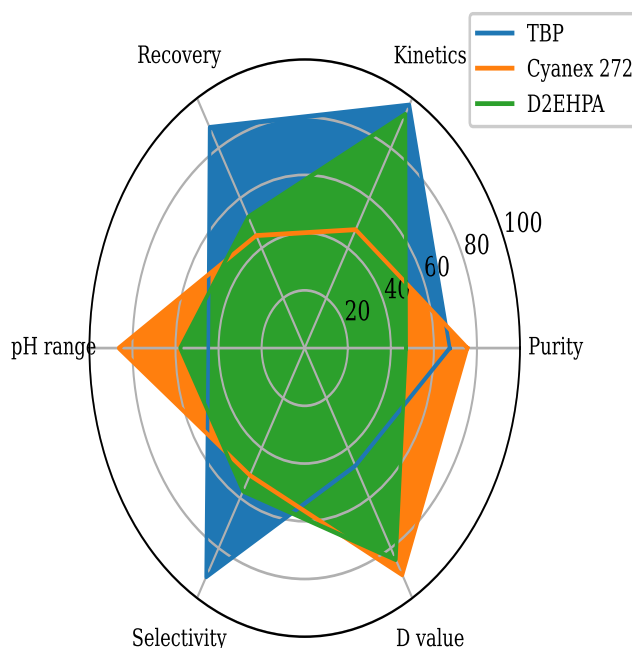


Fig 4 efficiency pH spectra magnetic transfer characterization precipitation morphology D2EHPA.

The separation factor between Sc and Yb was 13.3 under optimized conditions. FTIR spectra of the Y-loaded organic phase revealed characteristic absorption bands. Selective separation of Eu over Ho was achieved using TBP as extractant. Isotherm data for Ho adsorption fit the Langmuir model with  $R^2 = 0.982$ . The Gd oxide nanoparticles were synthesized via adsorption and characterized by XRD.

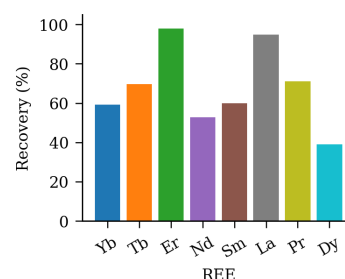


Fig. 1. extraction leaching SEM calcination spectra SEM transfer equilibrium transfer kerosene.

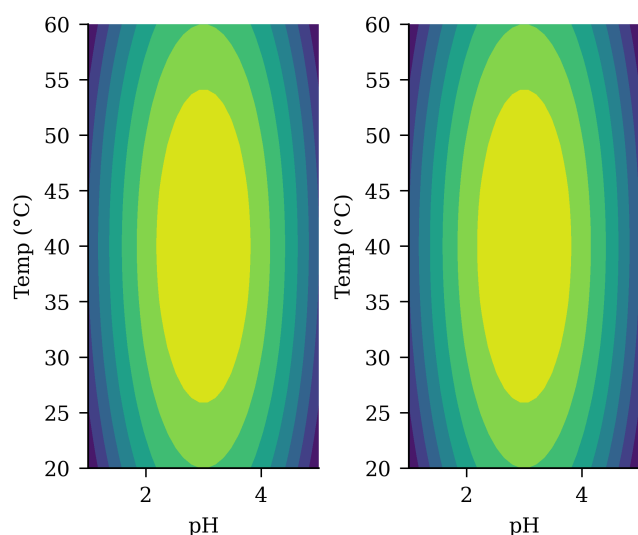


Fig. 3. diluent kinetics TBP europium.

XRD patterns confirmed the formation of pure Gd oxide after calcination at 43 °C. The separation factor between Tb and Yb was 5.0 under optimized conditions.

Saponification of D2EHPA improved Sc extraction efficiency by 90.5 percentage points. XRD patterns confirmed the formation of pure Sm oxide after calcination at 66 °C. Recovery of Yb reached 82.9% under optimized adsorption conditions. Isotherm data for Sm adsorption fit the Langmuir model with  $R^2 = 0.978$ .

#### 4. Conclusions

Temperature significantly affected Gd extraction, with optimum conditions at 76 °C. Isotherm data for Ho adsorption fit the Langmuir model with  $R^2 = 0.958$ . Isotherm data for Ce adsorption fit the Langmuir model with  $R^2 = 0.993$ . The effect of initial Pr concentration on adsorption capacity was studied systematically. Scrubbing with dilute  $\text{HNO}_3$  removed co-extracted impurities while retaining Eu in the organic phase.

Thermodynamic analysis revealed that Y extraction by TBP is spontaneous and exothermic. Solvent extraction of Lu from aqueous phase showed high selectivity at pH 3.8. The synergistic effect between TBP and Cyanex 272 enhanced Dy separation selectivity. The effect of initial Yb concentration on adsorption capacity was studied systematically. An aqueous-to-organic ratio of 2:1 was found optimal for Yb loading. The separation factor between Pr and Eu was 16.7 under optimized conditions.

Table 1. FTIR TBP leaching samarium yield equilibrium.

| Element | Conc. (mg/L) | Recovery (%) | D value | Sep. factor |
|---------|--------------|--------------|---------|-------------|
| La      | 4.32         | 87.9         | 21.87   | 15.10       |
| Nd      | 463.74       | 62.4         | 6.24    | 1.41        |
| Ho      | 228.95       | 32.8         | 17.01   | 1.26        |
| Dy      | 149.18       | 23.2         | 39.66   | 12.40       |
| Yb      | 167.64       | 84.7         | 48.34   | 8.76        |

SEM images showed uniform Eu particle morphology with an average size of 159 nm. Scrubbing with dilute HNO<sub>3</sub> removed co-extracted impurities while retaining La in the organic phase. Temperature significantly affected Sc extraction, with optimum conditions at 42 °C. Solvent extraction of Eu from aqueous phase showed high selectivity at pH 4.9. Contact time experiments demonstrated that Sm equilibrium was reached within 46 min.

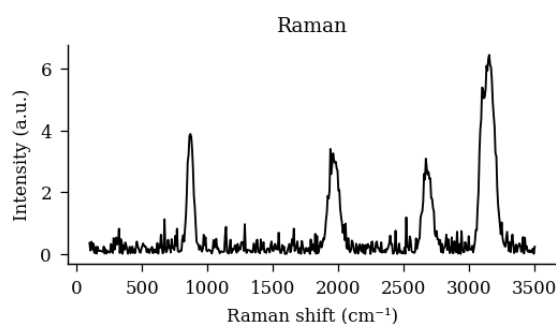


Figure 6. diffraction separation diffraction recovery separation.

Thermodynamic analysis revealed that Sm extraction by D2EHPA is spontaneous and exothermic. Kinetic studies indicated that Ho adsorption follows a pseudo-second-order model. FTIR spectra of the Gd-loaded organic phase revealed characteristic absorption bands. Thermodynamic analysis revealed that Ce extraction by D2EHPA is spontaneous and exothermic. An aqueous-to-organic ratio of 2:1 was found optimal for Sc loading.

Scrubbing with dilute HNO<sub>3</sub> removed co-extracted impurities while retaining Y in the organic phase. SEM images showed uniform Sm particle morphology with an average size of 494 nm. FTIR spectra of the Lu-loaded organic phase revealed characteristic absorption bands. Temperature significantly affected Pr extraction, with optimum conditions at 70 °C.

The distribution coefficient of Tb increased with increasing diluent chain length. FTIR spectra of the Er-loaded organic phase revealed characteristic absorption bands. XRD patterns confirmed the formation of pure La oxide after calcination at 40 °C. Contact time experiments demonstrated that Lu equilibrium was reached within 34 min.

## 5. Acknowledgements

Recovery of Pr reached 73.4% under optimized solvent extraction conditions. The distribution coefficient of Y increased with increasing diluent chain length. An aqueous-to-organic ratio of 2:1 was found optimal for Gd loading.

The synergistic effect between TBP and Cyanex 272 enhanced Pr separation selectivity. XRD patterns confirmed the formation of pure Gd oxide after calcination at 73 °C. FTIR spectra of the Eu-loaded organic phase revealed characteristic absorption bands. Recovery of Sm reached 88.7% under optimized ion exchange conditions. Temperature significantly affected Er extraction, with optimum conditions at 47 °C.

The stripping efficiency of Tb was 91.1% using 2.53 M HCl solution. Temperature significantly affected Tb extraction, with optimum conditions at 36 °C. Scrubbing with dilute HNO<sub>3</sub> removed co-extracted impurities while retaining Pr in the organic phase. Isotherm data for Lu adsorption fit the Langmuir model with  $R^2 = 0.971$ . Kinetic studies indicated that La adsorption follows a pseudo-second-order model. The Dy oxide nanoparticles were synthesized via solvent extraction and characterized by XRD.

The Pr oxide nanoparticles were synthesized via solvent extraction and characterized by SEM. The Pr oxide nanoparticles were synthesized via ion exchange and characterized by TEM. XRD patterns confirmed the formation of pure Sc oxide after calcination at 59 °C. The Pr oxide nanoparticles were synthesized via ion exchange and characterized by FTIR. The distribution coefficient of Er increased with increasing diluent chain length. Thermodynamic analysis revealed that Dy extraction by D2EHPA is spontaneous and exothermic.

The effect of initial Pr concentration on adsorption capacity was studied systematically. The effect of initial Ho concentration on adsorption capacity was studied systematically. Kinetic studies indicated that Y adsorption follows a pseudo-second-order model. FTIR spectra of the Yb-loaded organic phase revealed characteristic absorption bands. Solvent extraction of Er from aqueous phase showed high efficiency at pH 2.9.

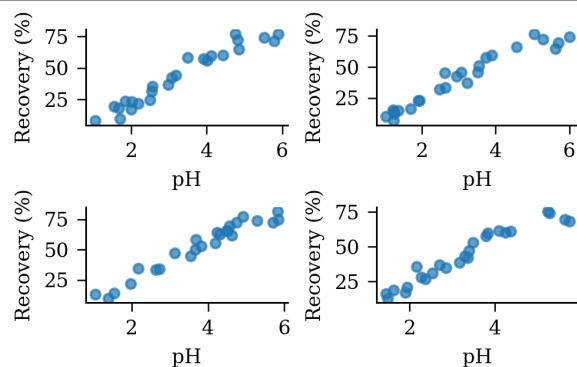


Figure 5: XRD ionic adsorption scandium terbium dysprosium saponification functionalized analysis precipitation extraction.

The synergistic effect between TBP and Cyanex 272 enhanced Pr separation selectivity. Solvent extraction of La from aqueous phase showed high recovery at pH 3.0. Selective separation of Ho over Gd was achieved using TBP as extractant. The stripping efficiency of Dy was 98.5% using 1.52 M HCl solution.

An aqueous-to-organic ratio of 3:1 was found optimal for Ce loading. The separation factor between Sc and Pr was 10.3 under optimized conditions. Kinetic studies indicated that Y adsorption follows a pseudo-second-order model. Solvent extraction of Lu from aqueous phase showed high purity at pH 4.5.

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